

AIMoodDiary: Hybrid Emotion Inference for Reflective Journaling and Mood Analytics

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Abstract—This paper presents an end-to-end affective journaling system that integrates transformer-based emotion classification, constrained LLM diary generation, and longitudinal mood analytics. The system supports free-text and keyword-based entries, optional speech-to-text input, and per-entry prediction storage for temporal pattern analysis. For emotion modeling, we compare baseline and transformer models on a 3-class mapping of GoEmotions (*negative, neutral, positive*) and deploy a RoBERTa 3-class model in the final pipeline.

We investigate whether keyword-guided hybrid inference yields more reliable emotion recognition than diary-only, LLM-generated content within reflective journaling systems. On a balanced held-out split ($n = 5,421$), RoBERTa achieves the strongest standalone performance (74.6% accuracy, 74.4% macro-F1). We then evaluate deployment behavior through a five-configuration ablation: full-text inference, keyword-only inference, diary-only inference, hybrid inference without calibration, and hybrid inference with calibration. Results show that keyword-only is strongest among diary-path variants (macro-F1 0.666), both 72/28 hybrid settings remain stable (macro-F1 ≈ 0.663), and diary-only underperforms (macro-F1 0.468) with a positive-skew tendency. Calibration is conservative, applying to 6.62% of samples and changing final labels in 0.24%. A fusion-weight sweep indicates a broad stable region, with best macro-F1 at 0.667 (0.90/0.10) and the deployed 72/28 setting near the upper plateau.

The system is positioned as a non-clinical reflective support platform and a practical testbed for responsible LLM-integrated journaling workflows.

Index Terms—Affective computing, emotion classification, reflective journaling, RoBERTa, large language models, hybrid sentiment inference, mood analytics

I. INTRODUCTION

Mental disorders remain a major global health challenge, while access to effective support remains uneven worldwide [1]. At the same time, global digital connectivity has expanded rapidly, creating opportunities for accessible tools that support emotional reflection without replacing professional mental health care [2].

Expressive writing is a well-established low-friction intervention associated with improvements in emotional processing and psychological well-being across clinical and non-clinical settings [3]. However, sustaining journaling behavior over time remains difficult in practice, and users vary in how quickly stable habits form [4]. This motivates systems that reduce user effort while maintaining meaningful reflective depth.

From a computational perspective, text-based emotion understanding is central to affect-aware systems. GoEmotions provides a large-scale benchmark for fine-grained emotion modeling [5]. Recent NLP practice shows that architecture

choice and training design strongly affect performance, motivating transformer-based approaches such as DistilBERT and RoBERTa [6], [7]. In parallel, large language models are increasingly explored in mental-health-adjacent workflows, but concerns about hallucination, reliability, privacy, and interpretability remain significant [8], [9].

Despite this progress, most prior work emphasizes either model benchmarking or standalone journaling interfaces; fewer studies evaluate a deployment-aligned pipeline that jointly combines constrained LLM generation, hybrid sentiment inference, and longitudinal analytics. Accordingly, this study investigates a focused research question: *can keyword-guided hybrid inference provide more reliable emotion recognition than diary-only, LLM-generated content within reflective journaling systems?* The novelty of this work lies not in a new classifier architecture, but in the integration and systematic evaluation of hybrid emotion inference, lexical calibration, and longitudinal analytics within a single deployable journaling workflow.

To keep terminology precise, we distinguish three related notions used throughout the paper. Fine-grained *emotions* from the source dataset are mapped into three *sentiment*-oriented categories (*negative, neutral, positive*) for deployment, and these per-entry sentiment predictions are subsequently aggregated over time into *mood*-trend analytics. We therefore use *emotion* for the underlying labels, *sentiment* for the deployed 3-class signal, and *mood* for its longitudinal interpretation.

Motivated by these challenges, we present AIMoodDiary, an end-to-end affective journaling system that integrates transformer-based sentiment modeling with structured LLM-assisted diary generation. The system supports free-text and keyword input, optional speech-based input via Whisper-based transcription [10], and longitudinal analytics over stored mood events. The deployed diary pipeline combines keyword and generated-diary predictions using 72/28 weighted fusion with lexical calibration to improve reliability in stress-sensitive contexts. On a balanced 3-class held-out split ($n = 5,421$), the deployed RoBERTa head reaches 74.6% accuracy and 74.4% macro-F1.

The main contributions of this work are as follows:

- Development of an end-to-end journaling pipeline that connects transformer-based emotion modeling with a deployable reflective workflow.
- A hybrid sentiment inference design combining user keywords and generated diary content via 72/28 weighted

fusion with lexical calibration.

- A reproducible evaluation framework with five-configuration ablation and fusion-weight sensitivity analysis on a fixed 3-class split.
- An interpretable longitudinal analytics layer for tracking mood trends, transitions, and engagement-relevant behavioral patterns over time.

The rest of the paper is structured as follows: Section II presents related work, Section III describes the methodology, Section IV reports experimental results, and Section V concludes with limitations and future work.

II. RELATED WORK

Emotion recognition from text has progressed rapidly with the availability of larger supervised datasets. A central milestone is GoEmotions, which provides fine-grained emotion labels at scale and highlights the difficulty of separating semantically adjacent emotions in realistic text [5]. Follow-up studies on GoEmotions continue to show that transformer-based methods improve performance over classical baselines, while still leaving substantial room for improvement on subtle classes [11]. Comparative analyses further show that preprocessing and modeling choices can materially change results, especially in noisy user-generated language [12].

Transformer architectures are now the dominant backbone for this task. While DistilBERT offers strong efficiency through distillation with lower computational cost [6], RoBERTa generally provides stronger predictive performance due to more extensive pretraining and optimization [7]. This trade-off highlights that deployment-oriented systems must balance inference quality, latency, and compute constraints rather than optimizing for benchmark scores alone.

Beyond classification accuracy, journaling and reflection literature provides the behavioral foundation for digital mood systems. Expressive writing has long been associated with meaningful psychological benefit under structured use [3]. In mobile settings, reviews of mood-monitoring applications report usability potential but also emphasize limited high-quality longitudinal evidence, particularly for younger populations [13]. Meta-analytic work on mental health and well-being apps similarly reports modest benefits with notable methodological variability across studies [14].

Recent work on large language models expands this space by enabling generated prompts, reflective dialogue, and context-sensitive text support. A systematic review finds strong promise but also recurring risks around hallucination, transparency, and governance in mental-health-adjacent uses [8]. A complementary risk-focused perspective stresses that LLM tools should support, not replace, professional care pathways [9]. Psychiatry-focused analyses reach similar conclusions and call for careful boundary setting, evaluation rigor, and responsible deployment practices [15].

Finally, emerging journaling systems and applied prototypes indicate that LLM-assisted reflection can improve perceived engagement and personalization when prompts are well constrained. Gamified journaling designs demonstrate how LLM

generation can support sustained interaction [16]. Comparative thesis work also suggests that LLM-supported emotional writing can produce different user experiences than standard journaling workflows [17]. Context-aware journaling studies further suggest that combining reflective writing with contextual signals can increase relevance and depth of reflection [18].

Decision-level fusion has been widely used in multimodal and ensemble settings to combine complementary predictive signals. Prior work shows that weighted late-fusion strategies can improve robustness when model components capture different aspects of the underlying sentiment signal [19]. These findings motivate hybrid inference strategies that combine complementary predictors instead of relying on a single source.

Most prior work still emphasizes either model benchmarking or standalone app experiences. Although prior studies have explored emotion classification, reflective journaling, and LLM-assisted interaction independently, limited research has examined how these components interact within a unified deployment-oriented system. This positions the present work as both an engineering artifact and a research testbed for practical, non-clinical digital self-reflection support. We found limited prior work that jointly evaluates constrained diary generation, hybrid fusion, calibration, and longitudinal analytics within one deployed reflective journaling pipeline.

III. METHODOLOGY

The proposed methodology connects three main components: (i) an offline emotion classification pipeline developed using the GoEmotions dataset, (ii) a deployment-aligned hybrid diary inference framework, and (iii) a reproducible ablation protocol designed to evaluate the same inference logic used in the production system. This overall structure follows prior research in emotion classification and transformer-based affective computing [5], [11], [12], while extending the workflow toward application-oriented LLM journaling and decision-level fusion strategies [16]–[19]. The deployed configuration is selected through empirical evaluation and then integrated into the production journaling workflow without changing the inference logic. Figure 1 presents the complete end-to-end methodology pipeline used in this study, including keyword extraction, diary generation, hybrid inference, calibration, and longitudinal analytics integration.

A. Data source, label mapping, and split design

The primary supervised resource used in this study is the GoEmotions dataset [5]. The publicly available train and test splits were used after preprocessing and label remapping. The original emotion labels were consolidated into an operational 3-class structure consisting of *negative*, *neutral*, and *positive* categories to align model training and deployment requirements.

For all ablation experiments and final evaluations, a balanced held-out test split was used ($n = 5,421$; 1,807 samples per class). Maintaining a balanced test distribution supports

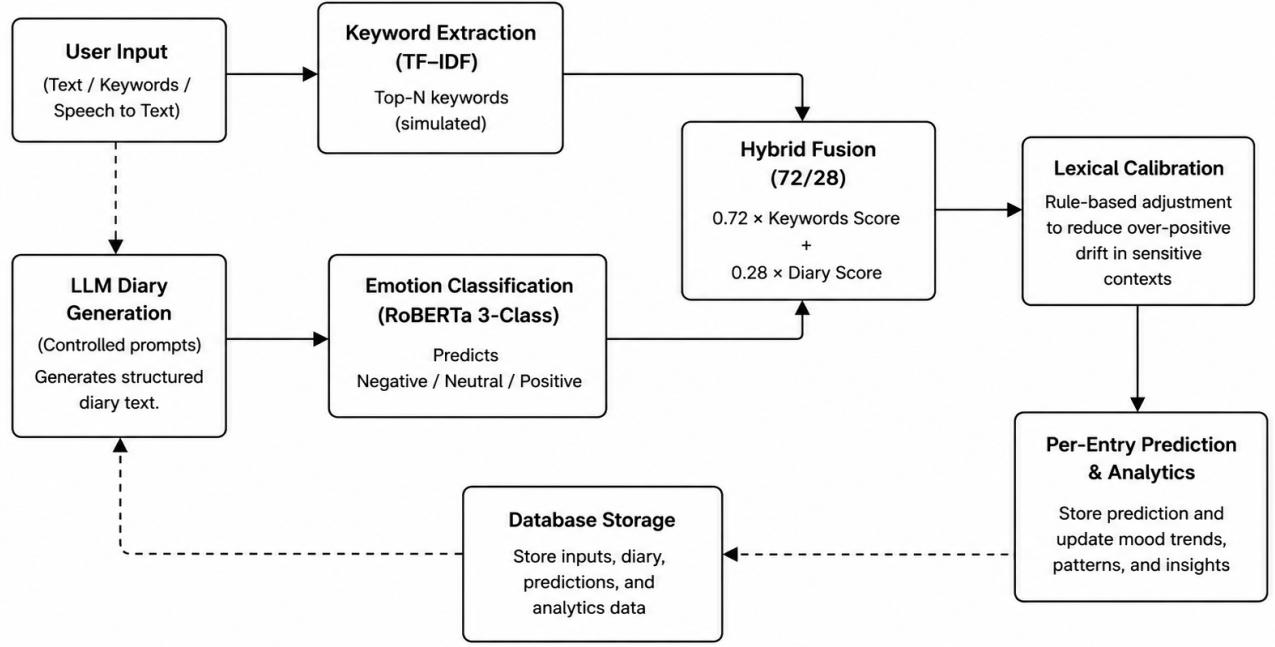


Fig. 1. End-to-end methodology and deployment pipeline of the proposed system. The workflow integrates keyword extraction, LLM diary generation, RoBERTa-based emotion classification, hybrid sentiment fusion, lexical calibration, and longitudinal analytics.

more stable macro-level evaluation and helps reduce interpretation bias caused by uneven class prevalence.

B. Model development pipeline

1) *Baseline and transformer families*: This study compares a lexical baseline model using TF-IDF with logistic regression, along with transformer-based architectures including DistilBERT and RoBERTa. The progression across these model families follows established NLP practices that evaluate compact transformer encoders against larger high-capacity architectures [6], [7]. The final deployed classifier is based on the RoBERTa 3-class architecture, implemented through the `EmotionClassifier` pipeline using fixed-length tokenization ($L_{\max} = 128$) and softmax-based probability outputs.

2) *Production parity and API integration*: A shared serving layer is used for both research evaluation and the deployed application, so that the inference logic measured in this study is identical to the one running in production. The backend exposes a direct text-based inference path and a full hybrid diary inference path, and optionally accepts speech input that is converted to text through Whisper-based transcription before entering the diary generation workflow [10].

3) *Diary generation and hybrid sentiment fusion*: Within the diary generation workflow, the LLM operates under keyword-constrained prompting so that generated diary content remains closely aligned with user-provided emotional cues. Sentiment inference is then performed on two separate inputs: (1) the raw user keywords and (2) the generated diary

text. The final sentiment probabilities are combined through fixed weighted averaging:

$$\mathbf{p}_{\text{hybrid}} = 0.72 \mathbf{p}_{\text{kw}} + 0.28 \mathbf{p}_{\text{diary}}.$$

The 72/28 weighting was not set arbitrarily but selected from the fusion-weight sensitivity analysis reported in Section IV. Preliminary experiments showed that strongly keyword-weighted settings provided a practical balance between classification stability and preservation of diary-generated reflective content, while yielding negligible benchmark loss relative to more extreme keyword-dominant configurations. The 72/28 point keeps the keyword signal dominant for stability yet retains a meaningful diary contribution to the reflective experience.

After fusion, an additional lexical post-processing calibration stage (`diary_sentiment_calibration.py`) applies rule-based probability adjustments designed to reduce over-positive sentiment drift in stress-leaning or emotionally sensitive contexts [20].

4) *Persistence and longitudinal analytics*: All mood entries are stored in PostgreSQL together with predicted labels, confidence scores, and per-class probability values. The analytics layer then processes this time-ordered history to generate trend analysis, weekday behavioral rhythms, emotional volatility patterns, transition tracking, and keyword-based summaries. This design separates immediate emotion prediction from longer-term reflective and longitudinal interpretation.

TABLE I
ABLATION PROTOCOL USED IN METHODOLOGY (SAME PIPELINE AS PRODUCTION DIARY ROUTE).

Mode	Description	Calibration
roberta_only	Full-text classifier baseline (Quick Check regime)	No
keywords_only	Classifier operating only on simulated keywords	No
diary_only	Classifier operating only on generated diary text	No
hybrid_no_calibration	72% weighted fusion of keyword and diary probabilities	No
hybrid_with_calibration	Complete production diary inference pipeline including calibration	Yes

C. Experimental protocol for research validation

To evaluate the behavior of the proposed pipeline beyond standalone classifier benchmarking, the study uses the experiments/ evaluation framework together with a frozen diary cache and a fixed held-out test split. Five different inference configurations are evaluated: `roberta_only`, `keywords_only`, `diary_only`, `hybrid_no_calibration`, and `hybrid_with_calibration`. To maintain deterministic and reproducible evaluation conditions, simulated keywords are generated using train-fitted TF-IDF top terms, allowing replayable keyword-based inference across all experiments.

In addition, a fusion-weight sensitivity analysis is performed using keyword weights of $\mathcal{W}_{kw} \in \{0.50, 0.60, 0.72, 0.80, 0.90\}$, where the diary contribution is defined as $1 - \mathcal{W}_{kw}$. This experiment is designed to measure how hybrid inference behavior changes as the pipeline becomes increasingly keyword-dominant.

D. Metrics, reporting, and statistical considerations

Primary reporting uses accuracy and macro-F1; weighted F1 and confusion matrices provide class-specific behavior analysis. Given class-sensitive evaluation requirements in multi-class settings, macro-level reporting is treated as first-order for cross-mode comparisons [21]. For future significance testing between paired models on the same test set, McNemar-style comparisons are recommended [22].

This methodology is intended for wellness journaling and reflective analytics, not clinical diagnosis or crisis intervention.

IV. RESULTS

A. Classifier benchmark on the 3-class split

We first evaluate the proposed emotion classification pipeline on the balanced GoEmotions 3-class test split consisting of negative, neutral, and positive labels ($n = 5,421$). The benchmark results are summarized in Table II. Among all evaluated approaches, the RoBERTa 3-class model achieved the strongest overall performance, reaching an accuracy of 74.64% and a macro-F1 score of 74.39%. This represents a substantial improvement over DistilBERT (64.06%

TABLE II
CLASSIFIER PERFORMANCE ON THE GoEMOTIONS 3-CLASS TEST SPLIT ($n = 5,421$).

Model	Accuracy	Macro-F1
Logistic + TF-IDF	0.537	0.556
DistilBERT	0.631	0.641
RoBERTa (7-class)	0.579	0.597
RoBERTa 3-class	0.746	0.744

TABLE III
RoBERTa 3-CLASS CONFUSION MATRIX (COUNTS). ROWS REPRESENT GROUND TRUTH LABELS AND COLUMNS REPRESENT PREDICTED LABELS.

	Neg.	Neu.	Pos.
True neg.	1,456	228	123
True neu.	407	1,118	282
True pos.	86	249	1,472

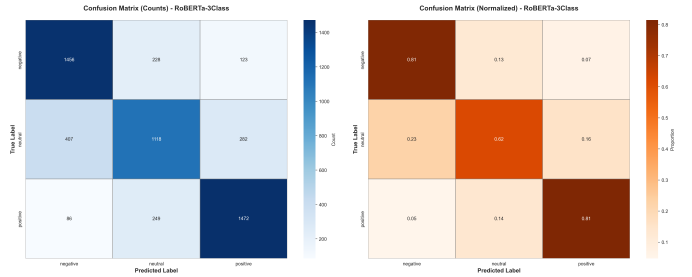


Fig. 2. Confusion matrix for the RoBERTa 3-class emotion classifier evaluated on the GoEmotions test split ($n = 5,421$).

macro-F1) and the classical logistic regression baseline using TF-IDF features (55.56% macro-F1).

The results further highlight the advantage of transformer-based contextual representations when handling short and emotionally ambiguous text samples. While the legacy RoBERTa 7-class configuration is included for historical reference, its label structure differs from the deployed 3-class setup and therefore should not be interpreted as directly comparable.

Further analysis of the RoBERTa 3-class confusion matrix (Table III, Fig. 2) shows that the neutral category remains the most challenging class to identify accurately, achieving a recall of 61.9%. A noticeable number of neutral samples are incorrectly predicted as either negative or positive (407 neutral→negative and 282 neutral→positive). This behavior is expected because neutral emotional expressions often share linguistic characteristics with both emotional extremes, particularly in short user-generated text.

In contrast, the model demonstrates stronger separation for the negative and positive classes, indicating that emotionally polarized expressions are easier to distinguish. These observations directly influenced the design of the deployed diary inference pipeline, where more conservative aggregation and calibration strategies were introduced to reduce unreliable sentiment transitions in borderline cases.

B. Hybrid diary inference ablation

To evaluate the deployed diary inference pipeline, we conducted an ablation study using five inference configurations on the same balanced 3-class test split. All experiments were performed using a frozen real diary cache (experiments/cache/diaries_real.jsonl) containing 5,421 generated diary entries. The evaluated configurations include: `roberta_only` (full input text), `keywords_only` (TF-IDF simulated top-8 keywords), `diary_only` (generated diary text only), `hybrid_no_calibration` (72/28 keyword-diary fusion), and `hybrid_with_calibration` (full production inference pipeline).

The results are summarized in Table IV and Fig. 3. Three important observations emerge from the analysis. First, the full-text RoBERTa configuration remains the strongest overall approach, achieving a macro-F1 score of 0.744. This result establishes the upper performance bound for the deployed sentiment pipeline and confirms the advantage of directly processing the complete user input.

Second, among the diary-path configurations, the `keywords_only` setup achieved the best performance with a macro-F1 score of 0.666, while both hybrid configurations produced nearly identical results (0.663 macro-F1). This indicates that the keyword representation preserves a large portion of the emotionally relevant information, while the addition of generated diary content contributes only marginally at the benchmark level.

Third, the `diary_only` configuration shows a substantial performance drop (macro-F1 0.468), accompanied by a strong positive prediction bias. In particular, the positive class achieved a recall of 88.1%, suggesting that LLM-generated diary text frequently softens or reframes negative emotional cues into more neutral or optimistic language. This behavior is consistent with known tendencies of generative language models to produce emotionally smoother outputs during text generation.

The calibration component remains intentionally lightweight within the production pipeline. At the 72/28 hybrid weighting, calibration was triggered for 359 out of 5,421 samples (6.62%), while modifying the final predicted label for only 13 samples (0.24%). Most label transitions occurred from `positive`→`neutral`, indicating that the calibration layer primarily acts as a conservative correction mechanism in borderline emotionally sensitive cases. Although calibration rarely changes the final predicted label, it provides a low-cost safeguard against over-positive sentiment drift in emotionally sensitive entries, where an incorrectly optimistic label is more harmful to a reflective user than a conservative neutral one. Its value is therefore behavioral and risk-oriented rather than reflected in aggregate accuracy.

Overall, these findings demonstrate that the proposed hybrid inference strategy operates as a deployment-oriented stabilization layer rather than a benchmark optimization technique.

TABLE IV
ABLATION STUDY ON THE FIXED GOEMOTIONS 3-CLASS TEST SPLIT ($n = 5,421$) USING A REAL DIARY CACHE AND SIMULATED KEYWORD EXTRACTION.

Configuration	Accuracy	Macro-F1
RoBERTa only (full text)	0.746	0.744
Keywords only	0.663	0.666
Hybrid 72/28, no calibration	0.661	0.663
Hybrid 72/28, + calibration	0.660	0.663
Diary only (LLM generated)	0.498	0.468

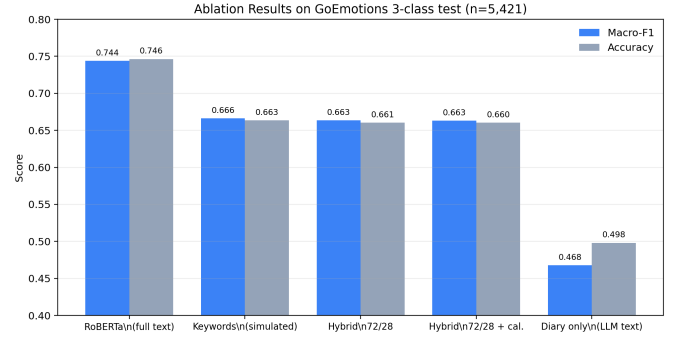


Fig. 3. Accuracy and macro-F1 comparison across five diary inference configurations.

TABLE V
CALIBRATION IMPACT ANALYSIS FOR THE 72/28 HYBRID INFERENCE CONFIGURATION.

Quantity	Value
Calibration applied	359 / 5,421 (6.62%)
Final-label flips (vs. no calibration)	13 / 5,421 (0.24%)
Dominant flip direction	positive → neutral (10 / 13)

While calibration has minimal impact on aggregate evaluation metrics, it improves behavioral consistency and reduces unreliable sentiment amplification during diary generation.

C. Error analysis

The observed limitations of the hybrid inference pipeline can largely be explained through two dominant error patterns. First, the ambiguity associated with the `neutral` class remains consistent across all evaluated configurations. For example, in the full-text RoBERTa setting, a substantial number of neutral samples are misclassified as either `negative` (407 cases) or `positive` (282 cases). This indicates that the hybrid fusion strategy cannot completely eliminate mid-valence overlap inherited from the underlying classifier.

Second, the generated diary text exhibits a noticeable polarity-smoothing effect. In the `diary_only` configuration, many originally negative and neutral samples shift toward the `positive` class (`negative`→`positive`: 938 samples; `neutral`→`positive`: 1,150 samples). This behavior contributes directly to the strong positive recall bias observed in Table IV and reflects the tendency of generative language

models to produce emotionally softened or optimistic narrative structures during text generation.

These findings further justify the deployment design choices used in the proposed system. In particular, the production pipeline maintains keyword-based sentiment signals as the dominant inference source, while the calibration layer operates primarily as a conservative correction mechanism rather than a global optimization strategy. This design helps reduce excessive sentiment amplification and improves stability in emotionally borderline cases.

D. Deployment relevance to the web application

The experimental findings directly correspond to the deployed web application and its production inference behavior. In particular, the live diary generation path uses the same `hybrid_with_calibration` pipeline evaluated in this study, including keyword-based classification, generated-diary classification, 72/28 weighted fusion, and lexical calibration. Consequently, the ablation and calibration results reported in this section reflect the actual inference behavior experienced by users in the deployed system rather than a separate offline evaluation pipeline.

From a deployment perspective, the results support the current operating configuration of the system. The keyword-dominant hybrid strategy provides substantially greater stability than diary-only inference, while the calibration layer introduces targeted corrections that help reduce over-positive sentiment drift without causing large-scale label instability. This behavior is particularly important in emotionally ambiguous or borderline cases, where excessive polarity amplification could negatively affect the consistency of generated diary outputs.

It is important to note that this paper focuses on model-level and pipeline-level evaluation rather than human-subject validation. Although the deployed application includes user-facing interfaces for journaling, analytics, and reflection, no formal user study is presented in the current work. Therefore, interface screenshots and deployment demonstrations are treated as implementation evidence rather than primary experimental validation.

Overall, these results demonstrate that the proposed inference architecture is suitable for real-world deployment settings where behavioral consistency, interpretability, and controlled sentiment generation are more important than benchmark optimization alone.

E. Fusion-weight sensitivity

To better understand the impact of keyword and diary contributions within the hybrid inference pipeline, we performed a sensitivity analysis across multiple keyword fusion weights $\{0.50, 0.60, 0.72, 0.80, 0.90\}$, where the diary contribution is defined as $1 - \mathcal{W}_{kw}$. Both calibrated and non-calibrated hybrid configurations were evaluated using the same frozen real diary cache.

The results are presented in Table VI and Fig. 4. A gradual improvement in macro-F1 can be observed as the keyword

TABLE VI
MACRO-F1 ACROSS KEYWORD/DIARY FUSION WEIGHTS USING THE REAL DIARY CACHE.

$\mathcal{W}_{kw}$	No calibration	+ Calibration
0.50	0.621	0.627
0.60	0.655	0.656
0.72	0.663	0.663
0.80	0.665	0.666
0.90	0.666	0.667

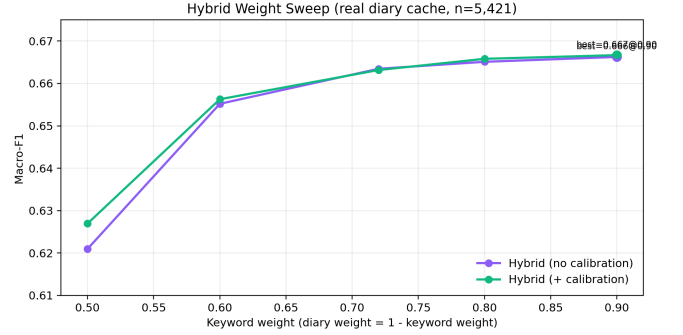


Fig. 4. Macro-F1 performance across different keyword/diary fusion weight configurations.

weight increases, indicating that keyword-based sentiment signals remain the most reliable source of emotional information within the deployed pipeline. This trend is consistent with earlier findings showing that generated diary text introduces polarity smoothing and positive sentiment drift.

The strongest overall result was obtained at the 0.90/0.10 fusion setting with calibration, achieving a macro-F1 score of 0.667. However, this improvement is marginal, representing only a +0.0003 increase compared to the `keywords_only` configuration (0.666 macro-F1). This suggests that heavily reducing diary contribution yields only limited benchmark gains while weakening the reflective role of generated diary content within the application.

The deployed 72/28 operating point achieves a macro-F1 score of 0.663 with calibration, placing it within the upper range of the sensitivity curve while still preserving a meaningful contribution from diary-generated text. From a deployment perspective, this operating point represents a practical tradeoff between classification stability and generative expressiveness, allowing the system to incorporate reflective diary content without excessively amplifying LLM-induced sentiment drift.

Overall, the sweep analysis demonstrates that the proposed hybrid architecture remains relatively stable across a broad range of fusion weights, while also confirming that moderate diary integration can be maintained without substantial degradation in aggregate performance.

V. CONCLUSION

This work presents AIMoodDiary, a non-clinical reflective journaling and wellness platform that integrates transformer-

based emotion classification, constrained LLM diary generation, and longitudinal mood analytics within a single deployable pipeline. The application is deployed as a publicly accessible web platform.

On the balanced 3-class GoEmotions test split ($n = 5,421$), the RoBERTa 3-class model remained the strongest standalone approach (74.6% accuracy, 74.4% macro-F1). Beyond standalone benchmarking, a controlled ablation of the deployed diary pipeline revealed a consistent ordering: full-text inference provides the upper performance bound, keyword-driven inference is the strongest diary-path source, and diary-only inference introduces a pronounced positive-sentiment bias. The keyword-dominant hybrid configuration remained stable across a broad range of fusion weights, while lexical calibration contributed targeted, risk-oriented corrections rather than aggregate accuracy gains.

The broader implication is that, for deployment-oriented reflective journaling, behavioral stability and controlled sentiment generation can matter more than marginal benchmark improvements. Anchoring predictions on user-provided keywords while treating generated diary text as a reflective rather than primary classification signal yields a workflow that is interpretable, reproducible, and resistant to LLM-induced positivity drift properties that are particularly valuable in emotionally sensitive contexts. The contribution of this work therefore extends beyond classifier performance alone, demonstrating an empirically evaluated inference workflow that links entry-level prediction to longitudinal, user-facing reflective insight.

AIMoodDiary is explicitly non-clinical and is not intended for diagnosis, treatment, or crisis intervention. The study also has clear limitations. Most importantly, user perception, usability, trust, and engagement were not evaluated: all results are model- and pipeline-level rather than human-subject findings, and remain future work. Further limitations include the use of simulated keyword inputs, social-domain training data, a single frozen LLM diary cache, and a benchmark-derived emotion taxonomy that may not fully reflect real-world journaling behavior. Future work will therefore prioritize human-centered evaluation including real-user keyword studies and prospective assessments of utility, trust, safety, and engagement within student populations alongside in-domain adaptation of the underlying classifier.

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